

# F24-067-D-TERRAVERSE 3D

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# Chapter 1

## 1. Introduction

The purpose of this project document is to outline a detailed plan for the development of an interactive virtual reality (VR) application that will transform video drone-footages of residential housing society into a 3D model for a virtual tour. The document will outline the project's problem we are going to address, what motivated us, scope of our project, deliverables, and all the technical processes involved in bringing an innovative solution to overcome the problem which we are going to implement in our final year project. The detailed overview of the project would mainly focus on creating a virtual environment of residential housing societies. This is a VR based application, which will provide a platform for stakeholders such as real estate builders associated with the society or society management to give a virtual tour of the society environment to the customers who are interested in investing in the society. A web application will be provided to the stakeholders, where they can upload video drone footage which will be used to generate a 3D model of the environment. The 3D model will be accessible through a VR based application using VR headset to virtually tour the environment. This project will help bridge the gap in the industry by providing an innovative solution to visualize and explore residential societies using modern-day technologies.

# Chapter 2

## 2. Project Vision

### 2.1 Problem Statement

The real estate market in Pakistan is growing at an unprecedented pace, with investors seeking safer, more Secure investments. Yet despite its rapid expansion, the industry still relies heavily on Traditional methods of Property visualization—photos, 2D maps, and occasional video tours. In traditional methods of exploring and assessing a new Residential society, individuals must physically visit each street and corner to get a comprehensive understanding of the area. This process is not only Time-consuming and physically demanding, but also impractical for those with Mobility limitations or busy schedules. The existing approaches to remotely visualize residential properties and societies, for instance, 'Google Street View and static 2D maps,' are inadequate to provide the required flexibility and interactivity for detailed exploration. Such models do not get updated in real-time and are very expensive in their development and maintenance; hence, not sufficient for the Customer, real estate developers, and investors who base their decisions on engaging, instructive visualizations. This indeed calls for a more efficient, interactive way of turning Drone footage into highly detailed, immersive 3D models. Increasingly, TerraVerse 3D meets this challenge with Virtual Reality and AI-driven technology to power the generation of 3D models from drone video footage. The system has been speeding up the whole process more economically and much more attractively.

## 2.2 Business Opportunity

### Educational and Training Purposes:

- **Architectural and Engineering Education:** Various universities and training institutes can use TerraVerse 3D to depict several real examples related to town planning and architectural designs.
- **Professional Training:** VR tours can be used in training by real estate professionals to understand the layout of the property and various sales techniques.

### Real Estate Marketing and Sales:

- **Virtual Property Tours:** Through virtual property tours, real estate agents and developers allow potential buyers to take a personal tour of their properties without having to go there physically. This ensures better and quicker decision-making and decreases the time a property spends on the market.
- **Improved Customer Interaction:** Meet customers in a more interactive manner, attracting more buyers to view properties and result in higher sales

### International Real Estate Investment:

- **Cross-Border Viewings:** TerraVerse 3D will be able to allow buyers and investors from any part of the world to view properties located in other regions virtually without having to physically visit the area. This will increase the market reach manifold for developers and real estate agents.
- **Enhanced Decision-Making:** By providing detailed and immersive virtual tours of properties, international investors can make more informed decisions about purchasing properties abroad. This can reduce the risk associated with investing in unfamiliar markets.

## 2.3 Objectives

- Generate 3D models from drone footage using Neural Radiance Fields (NeRF)
- Develop a VR environment compatible with Oculus Rift
- Implement an AI-powered Q&A system to provide information about the 3D society
- Optimize for real-time interaction and performance in VR



## 2.4 Project Scope

### 1. Data Collection and Preprocessing

- **Stabilizing and Preprocessing the Video Footage:**
  - Video stabilization software will be used to remove unwanted movements and jitters from footage.
  - Footage will be preprocessed by correcting lens distortions, adjusting lighting, and image quality will be enhanced for better 3D reconstruction.

### 2. 3D Model Reconstruction

- **Neural Radiance Fields (NeRF):**
  - **NeRF:** We will use neural networks to illustrate the model in 3D environment, allowing for highly detailed and realistic 3D reconstructions.
- **Optimizing the 3D Model for VR Performance:**
  - Simplifying the 3D model by reducing the polygon count while important details are being preserved.
  - Optimizing lighting and applying texture maps so that to enhance realism without compromising performance.
  - Ensuring that the model meets VR performance standards and providing a smooth and immersive experience.

### 3. VR Application Development

- **Building the VR Environment in Unity:**
  - The 3d model will be imported into unity
  - Setting the VR scene with lighting, textures, and interactive elements.
- **Implementing Smooth Navigation Controls:**
  - Developing navigation controls for users so that they can move around the VR environment.
  - Compatibility with Oculus Rift will be ensured including mappings and input handling.

## 4. AI and NLP Integration

- **Developing and training the AI Model:**
  - NLP technologies will be used to develop an Ai model which will be able to understand and respond to queries.
  - AI will answer questions like
    - Range of prices for plots of different blocks.
    - About where is commercial and residential area
    - Rules of society
    - Taxes by society

## 5. Testing and Feedback

- **Conducting User Testing Sessions:**
  - Organize sessions with target users to test the VR experience and AI interactions.
  - Gather qualitative and quantitative feedback on usability, performance, and overall user satisfaction.
- **Iterating on the Design and Functionality:**
  - Analyze user feedback to identify areas for improvement.
  - Iterate on the VR environment, AI interactions, and navigation controls to enhance the user experience.

## **Out-of-Scope Activities**

### **1. Real-time 3D Reconstruction**

- Only pre-recorded footage will be used to generate the 3d model. No real-time video processing and reconstruction.

### **2. Multi-Society Implementation**

- The initial implementation will be limited to one society. Scaling the solution to multiple societies is considered for future phases.

### **3. Extensive AI Training**

- The AI model will only provide information about the society. Or the information shared by society.

### **Model Updates:**

- The project will not update the 3d model of society once the model is created.

## 2.4 Constraints

### Technical Constraints:

- **Hardware Limitations:** The VR application requires users to have access to compatible VR headsets (e.g., Oculus Rift) and powerful enough hardware to run the application smoothly.

### User Constraints:

- **Accessibility:** Ensuring the VR application is easy-to-use and accessible to everyone with varying levels of technical expertise and physical abilities.

### Scalability Constraints:

- **Multiple Societies:** Initially, the project will focus on a single residential society, and expanding to multiple societies will require significant additional resources and planning. We will be covering 500 kanal of landmark.

### Real-time 3D Reconstruction:

- **Definition:** The whole project is based only on pre-recorded footage from drones to create the 3D model of the residential society. Real-time video processing and reconstruction is not implemented.
- **Explanation:** This limitation allows the project to focus on the accuracy and quality of the 3D models created from captured footage, instead of developing a complex system for real-time processing. By relying on pre-recorded footage, the project will create a more controlled environment for model generation, resulting in better overall quality. However, it constrains the application's ability to react to dynamic changes within the environment.

### **Multi-Society Implementation:**

We will be considering only one residential society for the initial phase of the project and will expand to multiple societies in the pipeline for upcoming phases.

### **Extensive AI Training:**

We will train NLP to provide information related to the specific residential society only, using the data that is shared by that society.

### **Model Updates:**

We will create only one version of the 3D model of the residential society and no further updates or revisions over time.

## **2.6 Targeted Users**

### **Property Managers and Real Estate Agents:**

- **Usage:** It will enable users to virtually expose already existing properties to potential renters or buyers. This will make multiple listings easier to handle.
- **Benefits:** Smoothens the process to view property without much need for physical visits and expands the ability to reach a larger audience.

### **Potential Homebuyers and Investors:**

- **Usage:** Browse properties and evaluate them from a distance with the best possible realistic view of your future potential investment without having to travel to it.
- **Benefits:** Offers convenience, efficiency, and better decision-making since buyers can experience properties in a virtual environment.

# Chapter 3

## 3. Software Requirement Specifications

### 3.1 List of features

- 3D Model Generation
- Interactive Virtual Reality Experience
- User-Friendly Web Application
- AI-Powered Q/A Information System
- Detailed Property Visualization

### 3.2. Functional Requirement

- The system should provide utilities to stabilize and pre-process the captured video output.
- It shall generate an accurate 3D model output from the uploaded drone footage.
- The system shall support the creation of a VR environment using Oculus.
- It shall be able to share related video drone footage with users via an intuitive web interface.
- The AI model can process user queries regarding the residential society and provide relevant answers, based on the data the model will receive.
- VR interaction shall include selecting points of interest and accessing further information on specific areas or amenities within the society.
- It shall ensure optimization in performance for 3D models in VR, with high navigability and low latency.

### 3.3. Quality Attributes

#### Availability:

- The system will have to reach a minimum of 99.99% uptime to ensure that the web application and VR features are always up and accessible for users and stakeholders alike.

#### Reliability:

- The system shall ensure that the latency in authenticating a user and establishing a network session is below 50 milliseconds to present a consistent response.

#### Usability:

- The interface shall be intuitive and self-descriptive so that users can easily learn how to move in the VR space and understand the application features with minimal training.

### 3.4. NON-FUNCTIONAL Requirements

#### Compatibility:

- Besides being compatible with major browsers like Chrome, Firefox, it should also be optimized to work on Edge, or desktop.

#### Performance:

- The system shall process uploaded drone footage and generate the 3D model within a maximum time of 30 minutes for a typical residential society (up to 500 kanal).
- The VR application shall maintain a frame rate of at least 60 frames per second (FPS) to ensure smooth navigation and an immersive experience.

#### User Support:

- The system shall provide a help and support feature, including FAQs, tutorials, and a ticketing system for user inquiries.

# Chapter 4

## 4. High Level Use Cases

### 4.1. Use Case Diagram

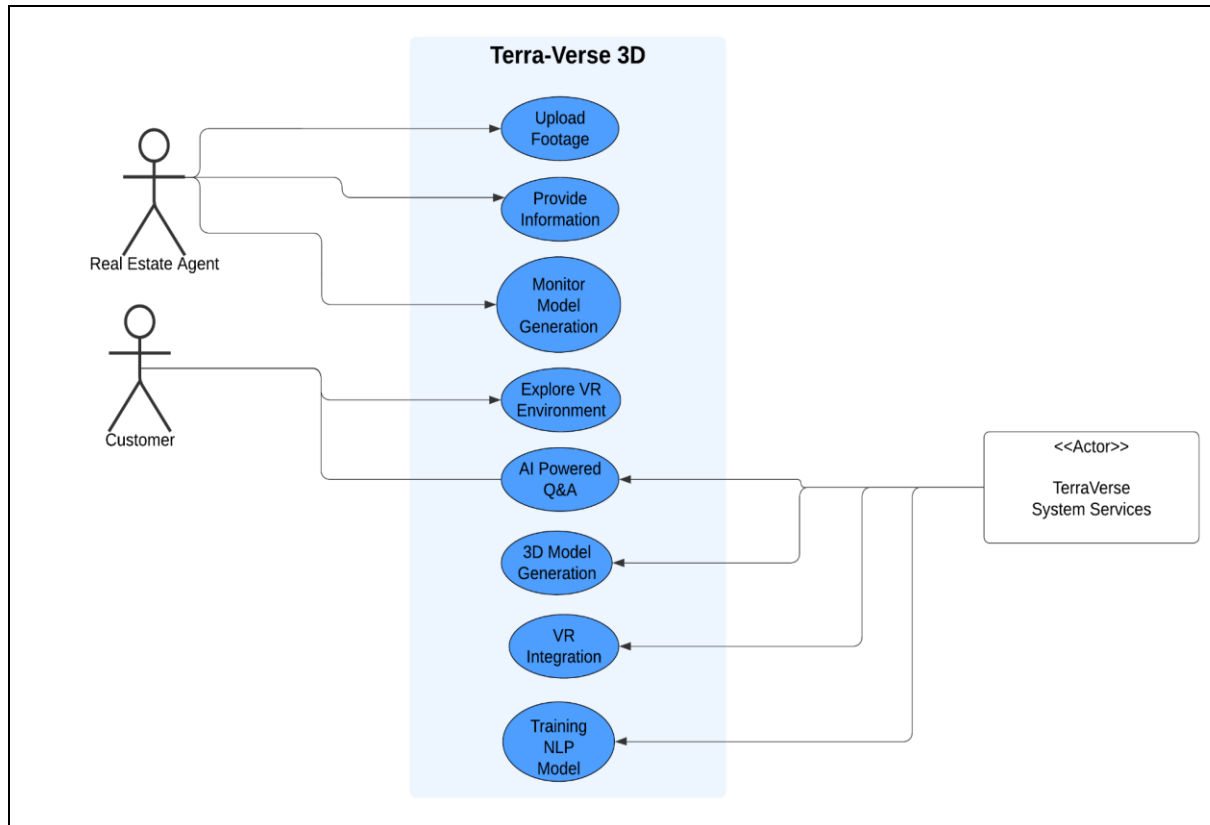


FIGURE 1: USE CASE DIAGRAM



# Chapter 5

## 5. Iteration Plan

### 5.1. Expanded Use Cases

TABLE 1: UC-1.1

|                                            |                                                                                                                                                                                  |                                                                                           |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Use Case ID:</b>                        | UC-1.1                                                                                                                                                                           |                                                                                           |
| <b>Use Case Name:</b>                      | Upload Footage                                                                                                                                                                   |                                                                                           |
| <b>Actors:</b>                             | Real Estate Agents                                                                                                                                                               |                                                                                           |
| <b>Description:</b>                        | User will upload drone footage of society                                                                                                                                        |                                                                                           |
| <b>Trigger:</b>                            | User wants to create a VR experience for society.                                                                                                                                |                                                                                           |
| <b>Preconditions:</b>                      | <ol style="list-style-type: none"><li>1. Open web application on browser</li><li>2. System should be logged in.</li><li>3. The user should be on the upload dashboard.</li></ol> |                                                                                           |
| <b>Post conditions:</b>                    | The video will be uploaded on the system.                                                                                                                                        |                                                                                           |
| <b>Normal Flow:</b>                        | <b>Actor</b><br><br>The user will select the upload option.                                                                                                                      | <b>System</b><br><br>The web application will present a dashboard with options to upload. |
| <b>Alternative Flows:</b>                  | N/A                                                                                                                                                                              |                                                                                           |
| <b>Exceptions:</b>                         | <ul style="list-style-type: none"><li>● Cannot upload footage due to unstable internet connection.</li><li>● Cannot connect to the cloud.</li></ul>                              |                                                                                           |
| <b>Frequency of Use:</b>                   | Whenever a user wants to select a new society.                                                                                                                                   |                                                                                           |
| <b>Specific Requirements:</b>              | <ul style="list-style-type: none"><li>● Stable internet connection.</li><li>● Video is in high quality graphics.</li></ul>                                                       |                                                                                           |
| <b>Technology and Data Variations List</b> | User Interface (UI) for inputting metadata.                                                                                                                                      |                                                                                           |

**TABLE 2: UC-1.2**

|                                            |                                                                                                                                                           |                                                                                                                                                                        |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID:</b>                        | UC - 1.2                                                                                                                                                  |                                                                                                                                                                        |
| <b>Use Case Name:</b>                      | Provide Information                                                                                                                                       |                                                                                                                                                                        |
| <b>Actors:</b>                             | Real Estate Agent                                                                                                                                         |                                                                                                                                                                        |
| <b>Description:</b>                        | User provides additional information about the footage, such as location and metadata.                                                                    |                                                                                                                                                                        |
| <b>Trigger:</b>                            | The user wants to add relevant details to the uploaded footage for better accuracy during 3D model generation.                                            |                                                                                                                                                                        |
| <b>Preconditions:</b>                      | Footage must already be uploaded.                                                                                                                         |                                                                                                                                                                        |
| <b>Post conditions:</b>                    | Information is successfully added and linked to the footage                                                                                               |                                                                                                                                                                        |
| <b>Normal Flow:</b>                        | <b>Actor</b> <ul style="list-style-type: none"> <li>● User selects the “Provide Information” option.</li> <li>● User inputs relevant metadata.</li> </ul> | <b>System</b> <ul style="list-style-type: none"> <li>● System prompts for the required information.</li> <li>● System validates and stores the information.</li> </ul> |
| <b>Alternative Flows:</b>                  | User skips adding information.                                                                                                                            |                                                                                                                                                                        |
| <b>Exceptions:</b>                         | Invalid or incomplete metadata                                                                                                                            |                                                                                                                                                                        |
| <b>Frequency of Use:</b>                   | Once per footage uploaded.                                                                                                                                |                                                                                                                                                                        |
| <b>Specific Requirements:</b>              | Active internet connection.                                                                                                                               |                                                                                                                                                                        |
| <b>Technology and Data Variations List</b> | User Interface (UI) for inputting metadata.                                                                                                               |                                                                                                                                                                        |

**TABLE 3: UC-1.3**

|                                            |                                                                                                               |                                                          |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>Use Case ID:</b>                        | UC 1.3                                                                                                        |                                                          |
| <b>Use Case Name:</b>                      | Monitor Model Generation                                                                                      |                                                          |
| <b>Actors:</b>                             | Real Estate Agents                                                                                            |                                                          |
| <b>Description:</b>                        | The user can see how much work is done on the generation process.                                             |                                                          |
| <b>Trigger:</b>                            | The user wants to know the time it will take to generate a model.                                             |                                                          |
| <b>Preconditions:</b>                      | The user has entered drone footage of society.                                                                |                                                          |
| <b>Post conditions:</b>                    | How much work is done, will be shown.                                                                         |                                                          |
| <b>Normal Flow:</b>                        | <b>Actor</b><br><br>Actor will open the dashboard.                                                            | <b>System</b><br><br>The system will show the work done. |
| <b>Alternative Flows:</b>                  | The system hasn't started working on generation.                                                              |                                                          |
| <b>Exceptions:</b>                         | Couldn't connect to the cloud or server.                                                                      |                                                          |
| <b>Frequency of Use:</b>                   | When model generation is in progress.                                                                         |                                                          |
| <b>Specific Requirements:</b>              | <ul style="list-style-type: none"> <li>● Drone footage is uploaded.</li> <li>● Cloud is connected.</li> </ul> |                                                          |
| <b>Technology and Data Variations List</b> | <ul style="list-style-type: none"> <li>● Unity</li> <li>● Nerf model for training</li> </ul>                  |                                                          |

**TABLE 4: UC-1.4**

|                                            |                                                                                                                                                                                      |                                                                                                                                                                         |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID:</b>                        | UC - 1.4                                                                                                                                                                             |                                                                                                                                                                         |
| <b>Use Case Name:</b>                      | Explore VR Environment                                                                                                                                                               |                                                                                                                                                                         |
| <b>Actors:</b>                             | Customer                                                                                                                                                                             |                                                                                                                                                                         |
| <b>Description:</b>                        | Customer explores the 3D model in a virtual reality environment.                                                                                                                     |                                                                                                                                                                         |
| <b>Trigger:</b>                            | User wants to explore the generated 3D model using VR                                                                                                                                |                                                                                                                                                                         |
| <b>Preconditions:</b>                      | A 3D model must be generated, and the VR environment should be ready.                                                                                                                |                                                                                                                                                                         |
| <b>Post conditions:</b>                    | Customers successfully navigate through the virtual society model.                                                                                                                   |                                                                                                                                                                         |
| <b>Normal Flow:</b>                        | <b>Actor</b> <ul style="list-style-type: none"> <li>● User selects “Explore VR Environment.”</li> <li>● User navigates and interacts with the 3D model using VR controls.</li> </ul> | <b>System</b> <ul style="list-style-type: none"> <li>● System loads the VR environment.</li> <li>● System allows the user to interact with various features.</li> </ul> |
| <b>Alternative Flows:</b>                  | N/A                                                                                                                                                                                  |                                                                                                                                                                         |
| <b>Exceptions:</b>                         | System fails to load the VR environment.                                                                                                                                             |                                                                                                                                                                         |
| <b>Frequency of Use:</b>                   | Multiple times for exploring the 3D model.                                                                                                                                           |                                                                                                                                                                         |
| <b>Specific Requirements:</b>              | VR headset (Oculus Rift) and compatible software.                                                                                                                                    |                                                                                                                                                                         |
| <b>Technology and Data Variations List</b> | Unity and Oculus Rift for VR integration.                                                                                                                                            |                                                                                                                                                                         |

**TABLE 5:UC-1.5**

|                                            |                                                                                                                                                            |                                                          |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>Use Case ID:</b>                        | UC - 1.5                                                                                                                                                   |                                                          |
| <b>Use Case Name:</b>                      | AI powered Q&A                                                                                                                                             |                                                          |
| <b>Actors:</b>                             | Client, System                                                                                                                                             |                                                          |
| <b>Description:</b>                        | The client can ask questions about society and the system will answer the questions.                                                                       |                                                          |
| <b>Trigger:</b>                            | Users are curious about society and want to know more.                                                                                                     |                                                          |
| <b>Preconditions:</b>                      | The system should be trained using data.                                                                                                                   |                                                          |
| <b>Post conditions:</b>                    | The user asks a question, and the system answers its question.                                                                                             |                                                          |
| <b>Normal Flow:</b>                        | <b>Actor</b><br><br>The user asks different questions about the society.                                                                                   | <b>System</b><br><br>The system answers those questions. |
| <b>Alternative Flows:</b>                  | The system couldn't answer the questions.                                                                                                                  |                                                          |
| <b>Exceptions:</b>                         | The system gives wrong answers.                                                                                                                            |                                                          |
| <b>Frequency of Use:</b>                   | Every time a client experiences the VR.                                                                                                                    |                                                          |
| <b>Specific Requirements:</b>              | <ul style="list-style-type: none"> <li>● Questions only related to society.</li> </ul>                                                                     |                                                          |
| <b>Technology and Data Variations List</b> | <ul style="list-style-type: none"> <li>● GPT-4 (or similar) as the NLP model.</li> <li>● AI training environment such as TensorFlow or PyTorch.</li> </ul> |                                                          |

**TABLE 6:UC-1.6**

|                                            |                                                                                                               |                                                                                                                                                                                           |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID:</b>                        | UC - 1.6                                                                                                      |                                                                                                                                                                                           |
| <b>Use Case Name:</b>                      | 3D Model Generation                                                                                           |                                                                                                                                                                                           |
| <b>Actors:</b>                             | System (NeRF Engine)                                                                                          |                                                                                                                                                                                           |
| <b>Description:</b>                        | System generates a 3D model from the uploaded drone footage.                                                  |                                                                                                                                                                                           |
| <b>Trigger:</b>                            | Footage has been uploaded, and the user wants to initiate 3D model generation.                                |                                                                                                                                                                                           |
| <b>Preconditions:</b>                      | Footage must be uploaded and validated.                                                                       |                                                                                                                                                                                           |
| <b>Post conditions:</b>                    | A 3D model is generated and ready for VR integration.                                                         |                                                                                                                                                                                           |
| <b>Normal Flow:</b>                        | <b>Actor</b> <ul style="list-style-type: none"> <li>● User initiates the model generation process.</li> </ul> | <b>System</b> <ul style="list-style-type: none"> <li>● System processes the footage using Neural Radiance Fields (NeRF).</li> <li>● System generates and refines the 3D model.</li> </ul> |
| <b>Alternative Flows:</b>                  | N/A                                                                                                           |                                                                                                                                                                                           |
| <b>Exceptions:</b>                         | System fails to generate a model due to insufficient data.                                                    |                                                                                                                                                                                           |
| <b>Frequency of Use:</b>                   | Once per footage set.                                                                                         |                                                                                                                                                                                           |
| <b>Specific Requirements:</b>              | Sufficient footage and processing power.                                                                      |                                                                                                                                                                                           |
| <b>Technology and Data Variations List</b> | NeRF model for 3D generation.                                                                                 |                                                                                                                                                                                           |

**TABLE 7: UC-1.7**

|                                            |                                                                                                                                   |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID:</b>                        | UC 1.7                                                                                                                            |
| <b>Use Case Name:</b>                      | VR Integration                                                                                                                    |
| <b>Actors:</b>                             | System                                                                                                                            |
| <b>Description:</b>                        | The system takes a 3D model and integrates into VR using an API.                                                                  |
| <b>Trigger:</b>                            | The client wants to experience the 3d model into VR.                                                                              |
| <b>Preconditions:</b>                      | 3D model generated.                                                                                                               |
| <b>Post conditions:</b>                    | A VR environment is created.                                                                                                      |
| <b>Normal Flow:</b>                        | <p style="text-align: center;"><b>System</b></p> <p>The system tak 3d model and integrates the model into the VR environment.</p> |
| <b>Alternative Flows:</b>                  | Non.                                                                                                                              |
| <b>Exceptions:</b>                         | The system fails to integrate the model.                                                                                          |
| <b>Frequency of Use:</b>                   | Whenever a user uploads data.                                                                                                     |
| <b>Specific Requirements:</b>              | Generated 3d model.                                                                                                               |
| <b>Technology and Data Variations List</b> | <ul style="list-style-type: none"> <li>● Python</li> <li>● Unity</li> </ul>                                                       |

**TABLE 8:UC-1.8**

|                               |                                                                                                                                                 |                                                                                                                                                                                                                                                   |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID:</b>           | UC - 1.8                                                                                                                                        |                                                                                                                                                                                                                                                   |
| <b>Use Case Name:</b>         | Training NLP Model                                                                                                                              |                                                                                                                                                                                                                                                   |
| <b>Actors:</b>                | System (AI)                                                                                                                                     |                                                                                                                                                                                                                                                   |
| <b>Description:</b>           | The system trains the Natural Language Processing (NLP) model using available data to enable the AI-powered Q&A feature in the VR environment.  |                                                                                                                                                                                                                                                   |
| <b>Trigger:</b>               | The AI system is ready for training with the data provided by the user.                                                                         |                                                                                                                                                                                                                                                   |
| <b>Preconditions:</b>         | Adequate training data must be available and labeled for the NLP model to process.                                                              |                                                                                                                                                                                                                                                   |
| <b>Post conditions:</b>       | The NLP model is trained and ready to provide answers to user questions in the VR environment.                                                  |                                                                                                                                                                                                                                                   |
| <b>Normal Flow:</b>           | <b>Actor</b> <ul style="list-style-type: none"> <li>● System administrator initiates the NLP model training.</li> </ul>                         | <b>System</b> <ul style="list-style-type: none"> <li>● The system processes the training data.</li> <li>● The model is trained and tested for accuracy.</li> <li>● The trained model is integrated into the AI-powered Q&amp;A system.</li> </ul> |
| <b>Alternative Flows:</b>     | The system fails to train the model due to inadequate data or a processing error.                                                               |                                                                                                                                                                                                                                                   |
| <b>Exceptions:</b>            | Insufficient training data or issues with model convergence.                                                                                    |                                                                                                                                                                                                                                                   |
| <b>Frequency of Use:</b>      | Once when training the initial AI model or for updates.                                                                                         |                                                                                                                                                                                                                                                   |
| <b>Specific Requirements:</b> | <ul style="list-style-type: none"> <li>● Adequate labeled data for model training.</li> <li>● High processing power for AI training.</li> </ul> |                                                                                                                                                                                                                                                   |



|                                                |                                                                                                                                                         |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Technology and Data<br/>Variations List</b> | <ul style="list-style-type: none"><li>• GPT-4 (or similar) as the NLP model.</li><li>• AI training environment such as TensorFlow or PyTorch.</li></ul> |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|

## 5.2. Activity Diagram

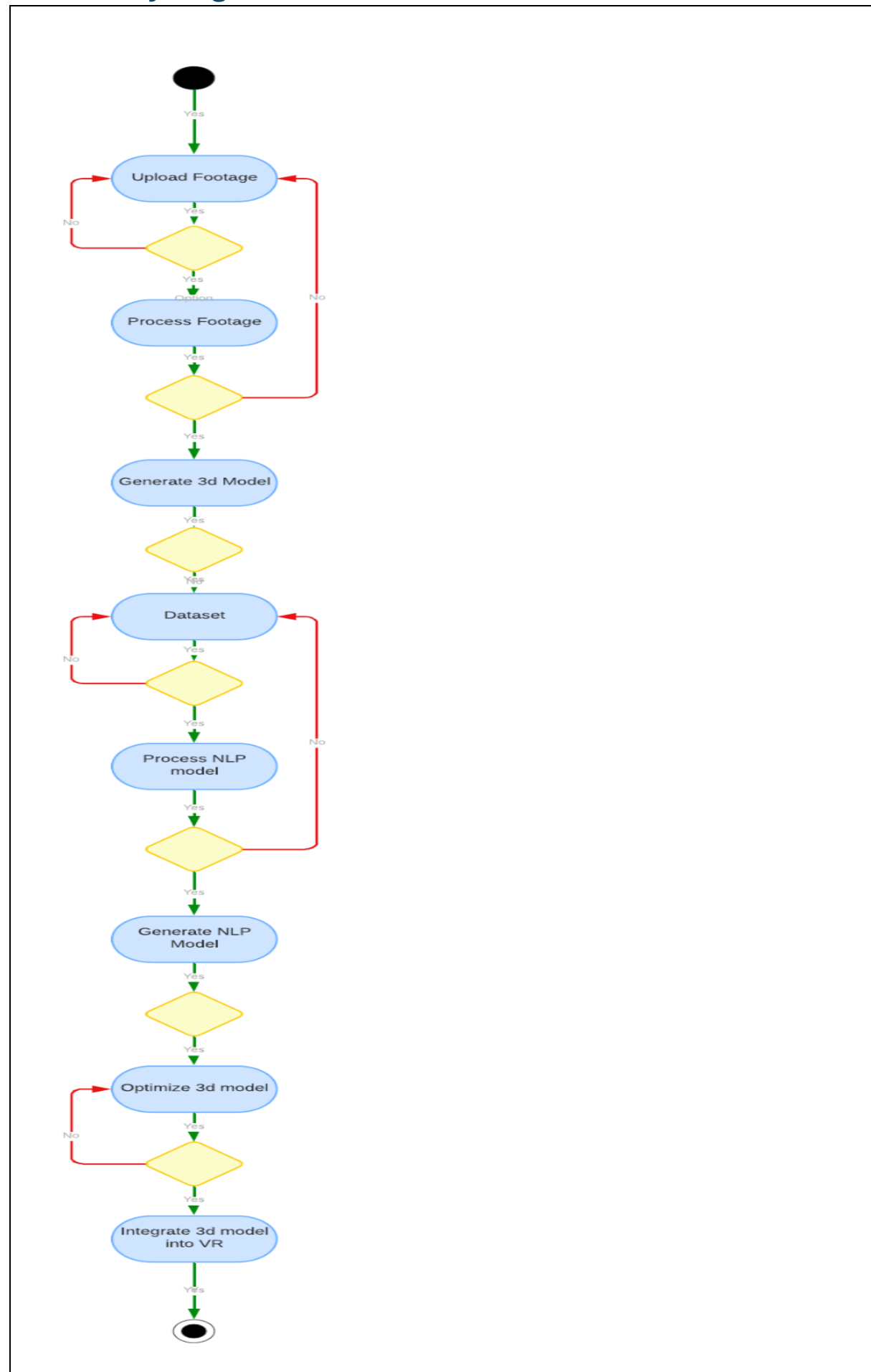


FIGURE 2: ACTIVITY DIAGRAM

## 5.3. Domain Diagram

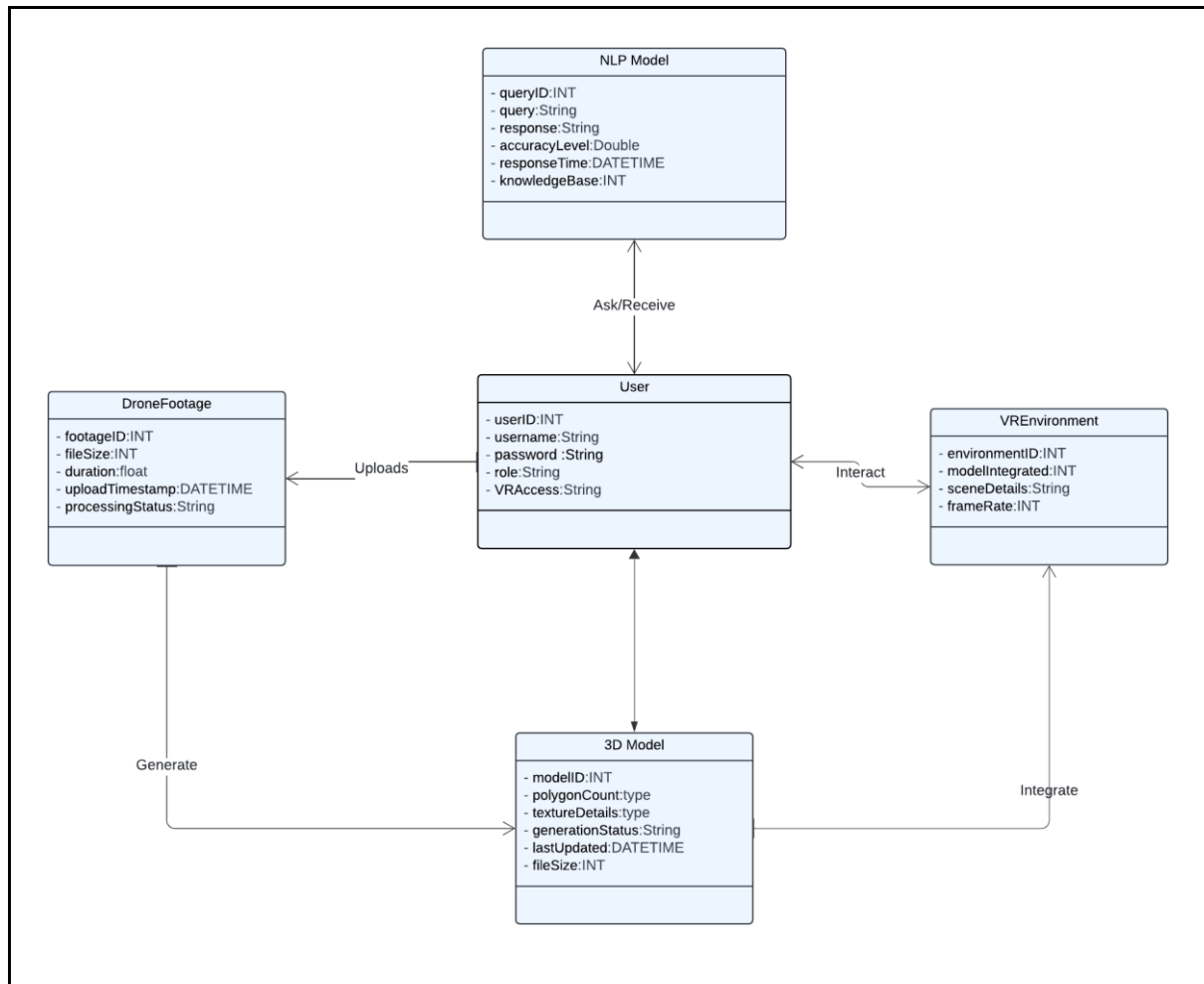


FIGURE 3: DOMAIN DIAGRAM

## 5.4. System Sequence Diagram

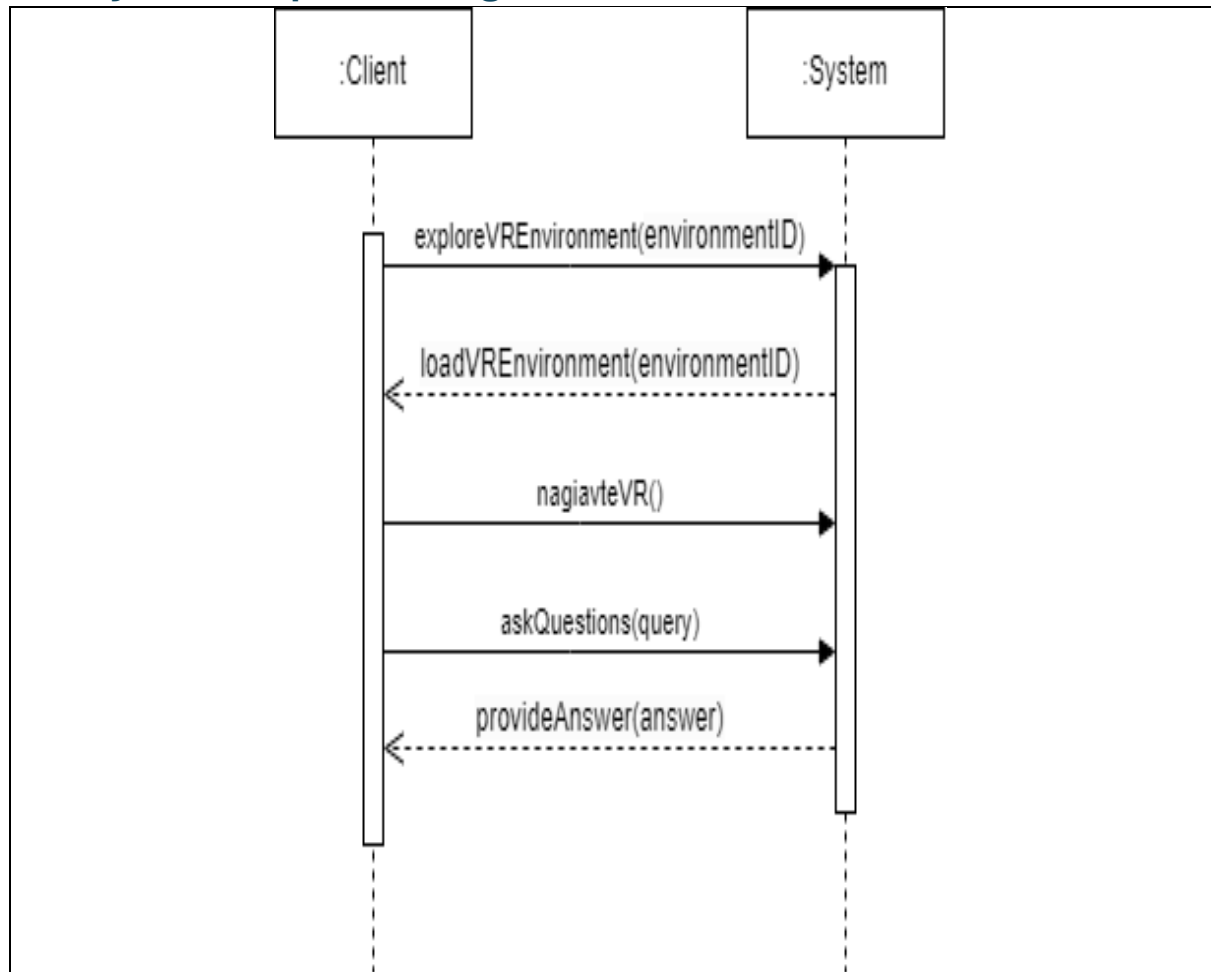


FIGURE 4: SYSTEM SEQUENCE DIAGRAM 1

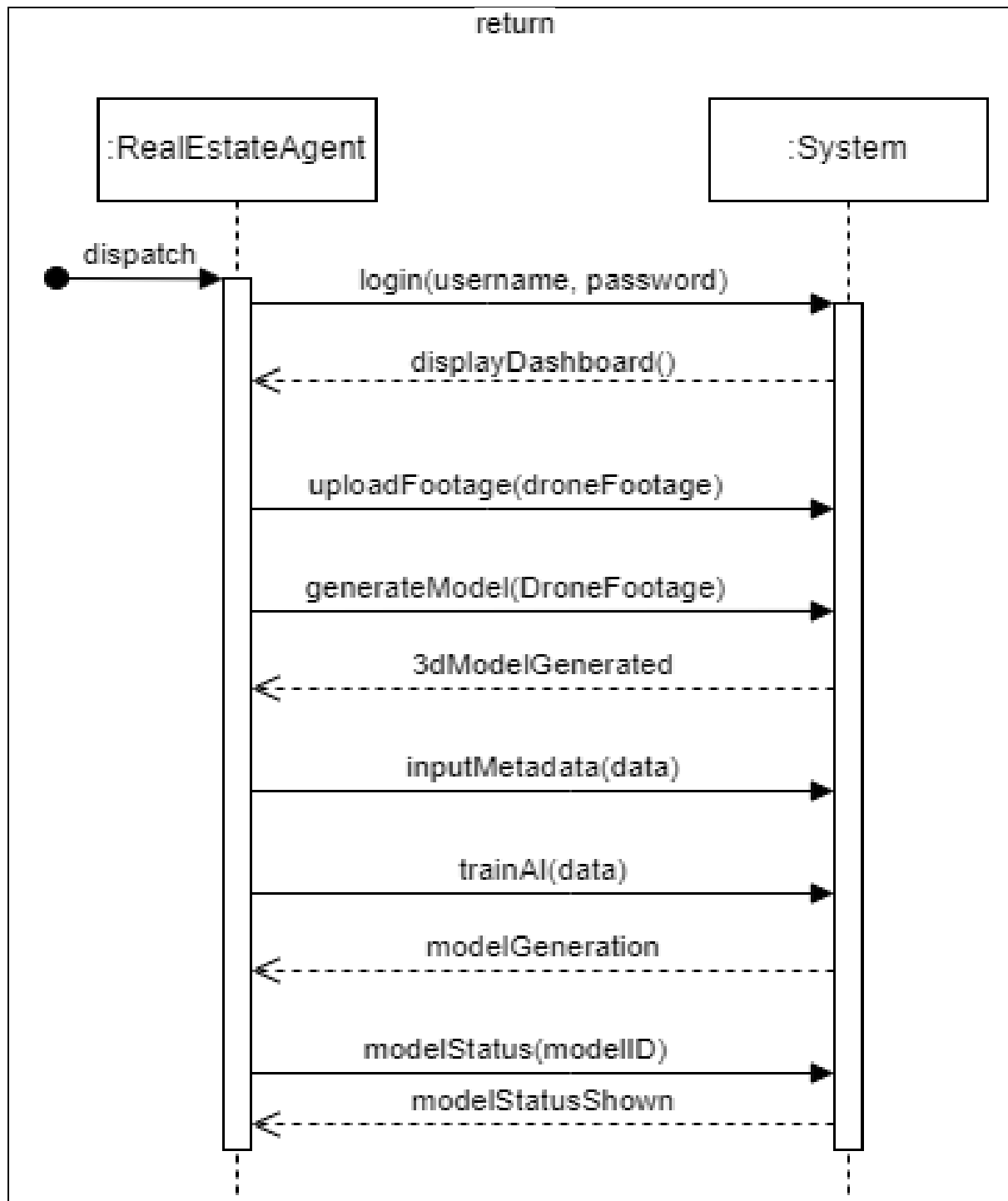


FIGURE 5: SYSTEM SEQUENCE DIAGRAM 2

## 5.5. Operation Contracts

### Contract CO1:

**Operation:** Upload Footage

**Responsibility:** User uploads drone footage of the society.

**Type:** System

**Pre-Conditions:**

- The web application is open in a browser.
- The user is logged in.
- The user is on the upload dashboard.

**Post-Conditions:** The video is successfully uploaded to the system.

### Contract CO2:

**Operation:** Provide Information

**Responsibility:** User provides additional information about the footage, such as location and metadata.

**Type:** System

**Pre-Conditions:** Footage must already be uploaded.

**Post-Conditions:** Information is successfully added and linked to the footage.

### Contract CO3:

**Operation:** Monitor Model Generation

**Responsibility:** User monitors the progress of the 3D model generation process.

**Type:** System

**Pre-Conditions:** The user has entered drone footage of the society, and the model generation process has started.

**Post-Conditions:** The system is going to display the progress of the model generation.

#### **Contract CO4:**

**Operation:** Explore VR Environment

**Responsibility:** The user will explore and interact with the 3D model that is created in a virtual reality environment.

**Type:** System

**Pre-Conditions:** A 3D model is generated, VR environment is prepared, and the user should possess a VR headset (e.g., Oculus Rift) with the necessary software.

**Post-Conditions:** The customer can successfully navigate a 3D model of the virtual society.

#### **Contract CO5:**

**Operation:** AI-Powered Q&A

**Responsibility:** The client can ask questions about the society, and the system will be able to answer the question.

**Type:** System

**Pre-Conditions:** The system must be trained with data, and the 3D model and VR environment should also be ready.

**Post-Conditions:** The user will be able to ask a question, and the system will provide a correct answer.

#### **Contract CO6:**

**Operation:** 3D Model Generation

**Responsibility:** The system is going to generate a 3D model from an uploaded drone footage.

**Type:** System

**Pre-Conditions:** Validated footage, and adequate processing power must be available.

**Post-Conditions:** A 3D model will be generated and ready to be integrated into the VR environment.

### **Contract C07:**

**Operation:** VR Integration

**Responsibility:** The system will take a 3D model and integrate the model into a Virtual Reality environment using the API.

**Type:** System

**Pre-Conditions:** Generated 3D model and should be available.

**Post-Conditions:** A Virtual Reality environment is created with the integrated 3D model.

### **Contract C08:**

**Operation:** Training NLP Model

**Responsibility:** The system uses available data to train the NLP model, enabling the AI-powered Q&A feature in the Virtual Reality environment.

**Type:** System

**Pre-Conditions:** Adequate data must be available and labeled for the NLP model to process.

**Post-Conditions:** The NLP model is trained and ready to provide answers to user questions within the VR environment.



## 5.6. Class Diagram

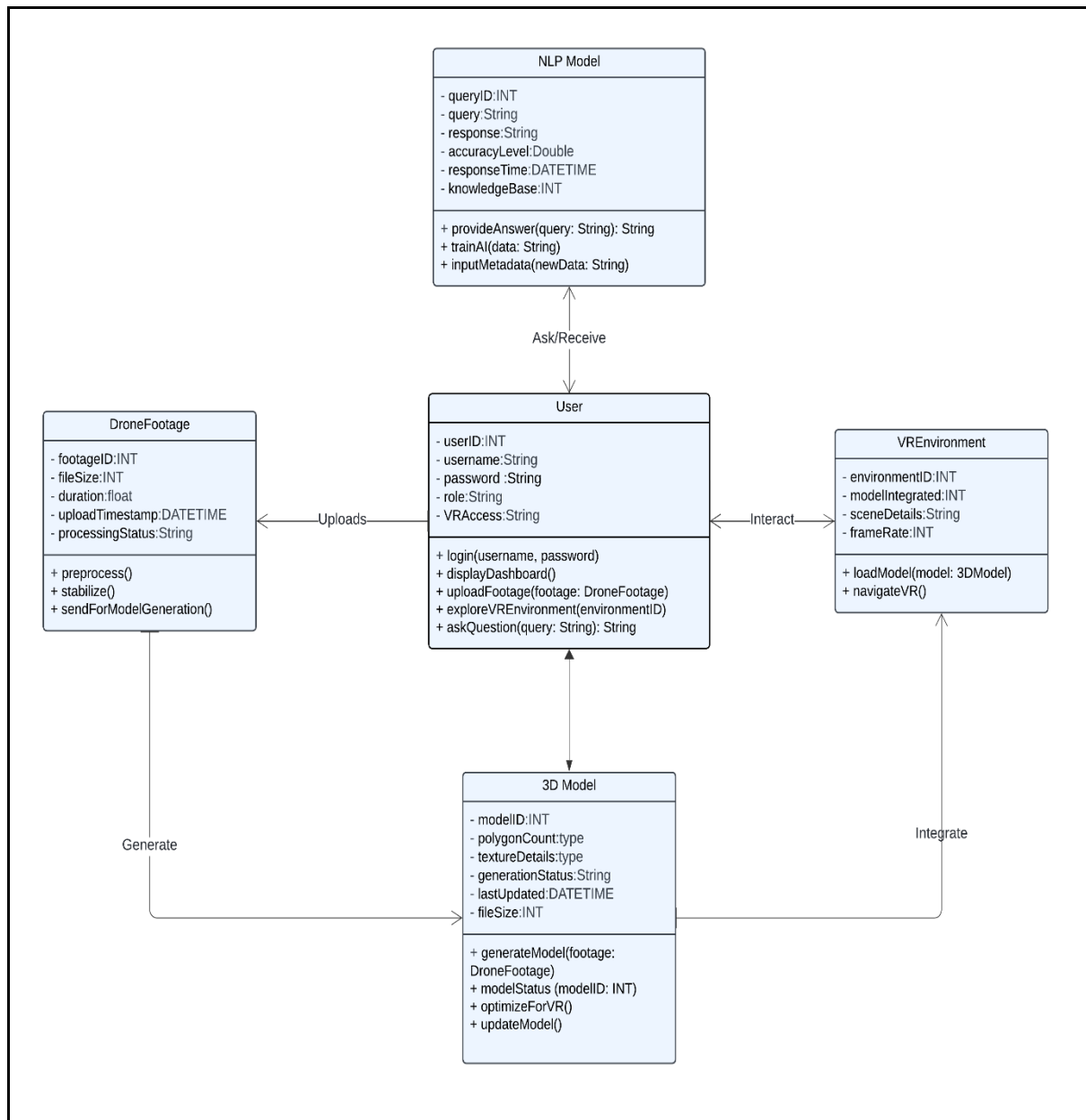


FIGURE 6: CLASS DIAGRAM

## 5.7. Architecture Diagram

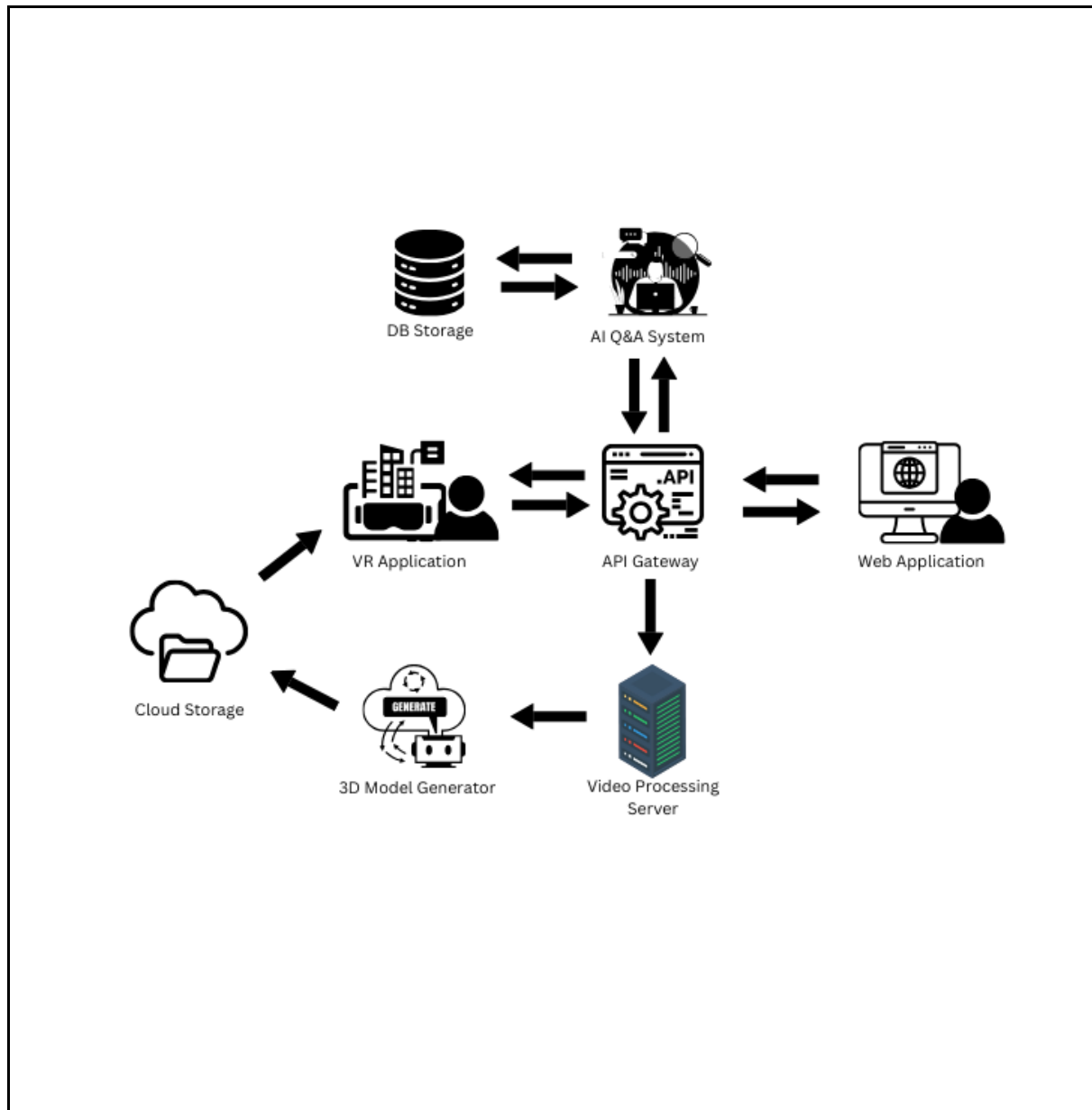


FIGURE 7: ARCHITECTURE DIAGRAM

## 5.8. Deployment Diagram

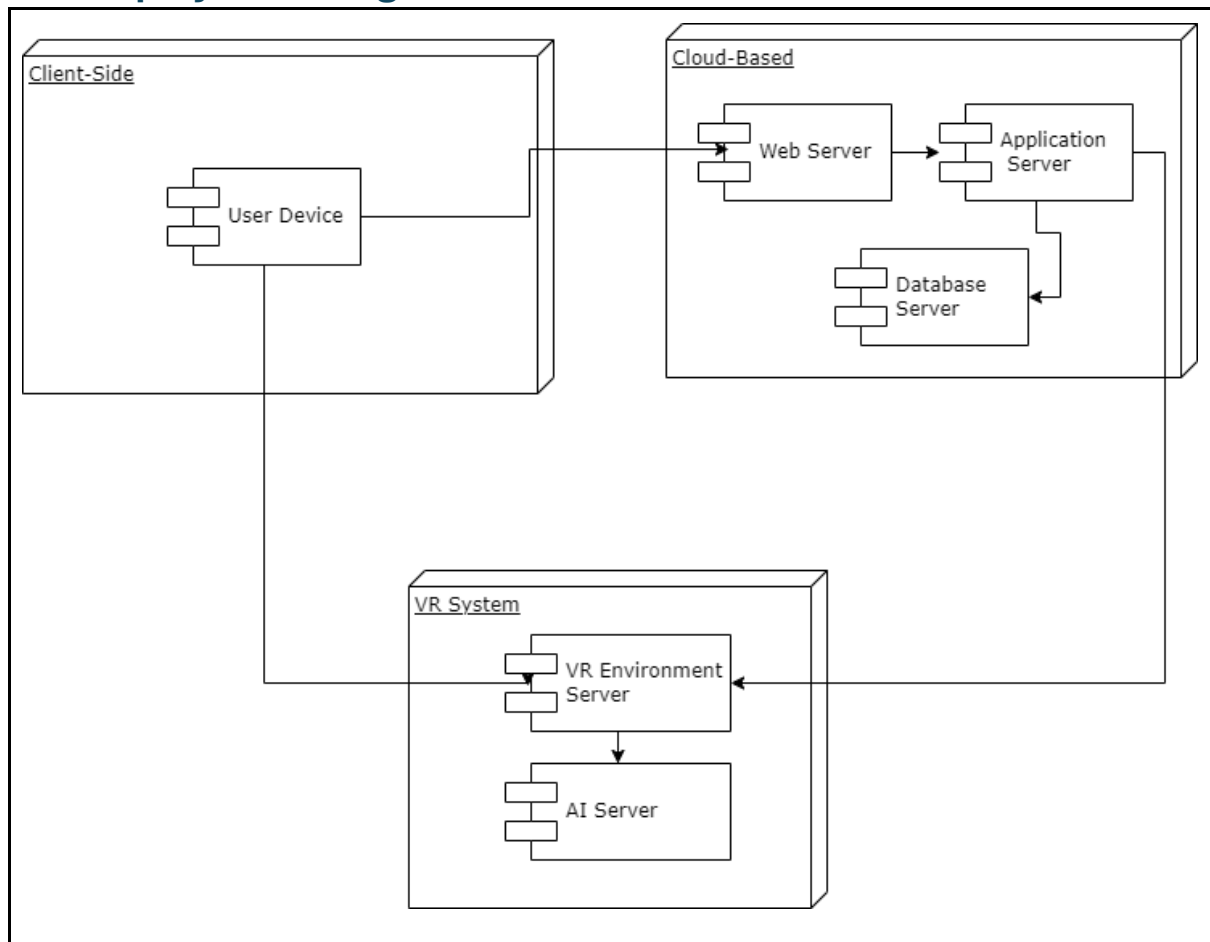


FIGURE 8: DEPLOYMENT DIAGRAM

## 5.9. Package Diagram

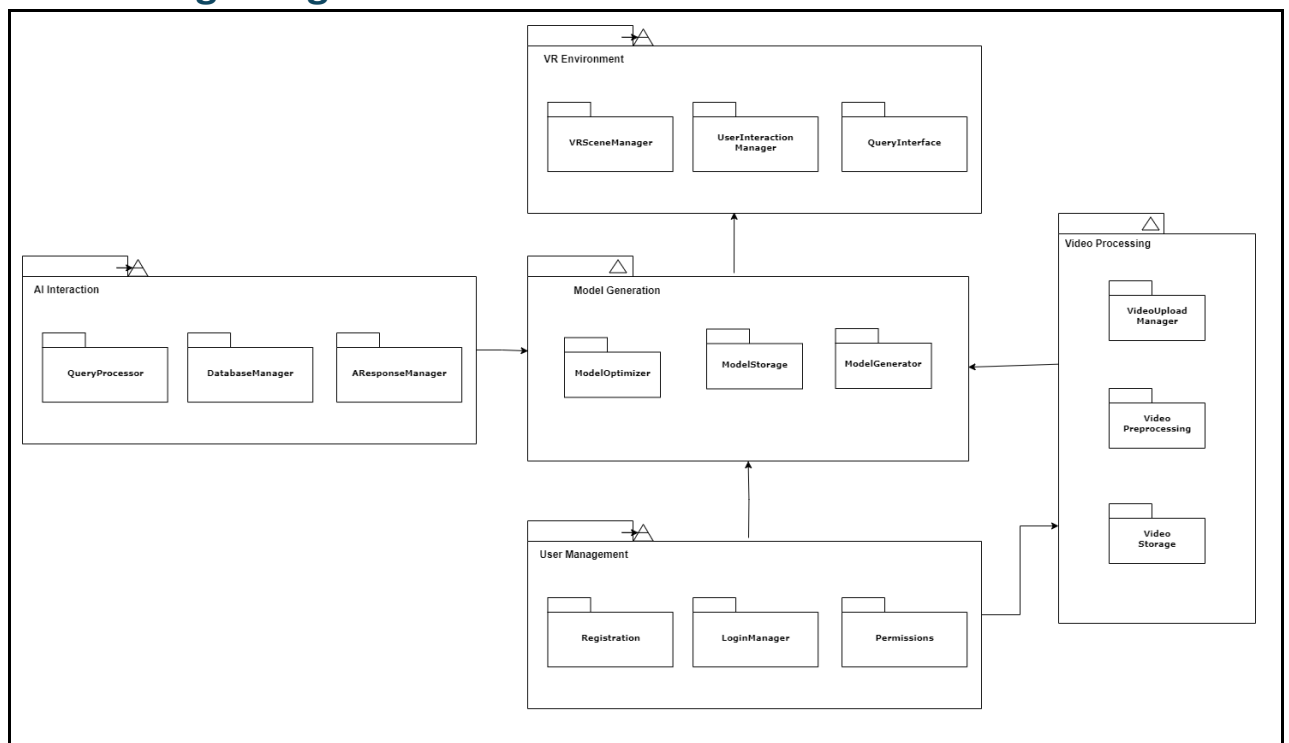


FIGURE 9: PACKAGE DIAGRAM

## Conclusion

The TerraVerse 3D project manages to effectively combine the most advanced technologies, enabling a revolutionary change in residential property visualization. This endeavor, with VR, AI, and cutting-edge 3D model generation techniques, allows amazing levels of immersive-ness, interactivity, speed, and ultimately a better experience for the real estate players in question. With this, it mitigates the drawbacks presented by traditional visualization methods and presents a fresh approach. This way, the user's experience and decision-making processes are extremely enriched. It forms, in a way, a new standard for the industry-classic showcase of future developments and scalability. It is approached with clarity of objectives, heavy planning, and iterative improvements based on user feedback. Indeed, this is really a big move toward transforming real estate marketing and exploration.